



STINK3114 NATURAL LANGUAGE PROCESSING

Semester II Session 2025/2026 (A252)

Assignment 1: Literature Review

Using Language Models for assessment of users' satisfaction with their partner in Malaysia

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Introduction

Background & Context

NLP has totally changed the way computers understand and interpret human language. In NLP, sentiment analysis is a key method that is applied to handle text with the goal of categorising it, either as positive or negative or as neutral. Over the past few years, social media platforms have turned into critical online spaces where people can share their everyday reflections, hassles, and feelings freely. As a result, such sites can make excellent real-time archives of the study of such complex processes in society, such as the well-being and fulfillment of romantic relationships. Knowledge of domestic well-being and satisfaction of partners is vital in determining the overall mental health and social stability of a society. Nevertheless, such measures are extremely ineffective and costly to assess using conventional psychological testing, since they require a lot of time, capital, and human resources.

Problem Statement

Although NLP is a promising computational substitute to conventional surveys, the application of these methodologies to the Malaysian online environment poses a major technical challenge. The language mostly used by the Malay social media population is extremely informal, with distinctive local slang, short forms and heavy code-mixing of Malay and English (often called Bahasa Rojak). Conventional machine learning systems that are based on strict lexicon policies cannot handle this unstructured and noisy data, and do not reflect the subtle emotional nuances of intimate human relationships. Moreover, Malay does not have large, high-quality annotated digital resources, as compared to high-resource languages such as English. It is so under-resource that it is extremely hard to train and optimise high-quality sentiment analysis models that can grasp the rich emotional syntax of the relationship dynamics in Malaysian contexts.

Research Gap

Although there have been recent developments in deep learning with low-resource languages, there is a significant gap in the literature. Most of the existing Malay NLP studies are confined to publicly/generally oriented topics, including mental health awareness, political sentiment or product reviews. A strong lack of literature using NLP in the very personal, interpersonal areas of life such as marriage and dating is also evident. Although pioneering applications in other low-resource languages, e.g. Persian (Habibzadeh and Asadpour, 2025), have effectively applied pre-trained language models to predict life partner satisfaction, this methodology has not been experimented in Malaysia. At present, no labelled dataset or specialised language model exists that can be used to assess relationship satisfaction in the context of Malaysian social media, which is code-mixed.

Research Objectives

To address the identified research gap, this study aims to achieve the following specific and measurable objectives:

- **To construct** a labelled dataset of informal, code-mixed Malaysian social media text specifically focused on romantic relationships and partner satisfaction.
- **To evaluate** the performance of pre-trained language models against traditional machine learning classifiers when processing noisy, low-resource Malay-English text.
- **To develop** an optimised sentiment analysis model capable of accurately categorising Malaysian relationship satisfaction into positive, negative, and neutral sentiments.

Motivation / Contribution

The pressing need to have scalable, cost-effective tools used to monitor social well-being is very motivating in this research. In practice, an NLP model that is localized to the linguistic peculiarities of the area will offer sociologists, policymakers and support groups in the community a fast, automated approach to assess relationship health as opposed to the traditional methods that are slow and resource intensive. This research will also contribute immensely to the existing literature on low-resource NLP by providing a new, domain-specific code-mixed Malay dataset. Additionally, it will provide a standard of using advanced language models to social problems with a private domain, extending the variation of what sentiment analysis can perform in Southeast Asian linguistic settings.